# pve-fcsan: How It Works

Version 1.8.0

## Goals and the stack mapping

pvesan rebuilds the vCenter SAN workflow on top of parts Proxmox VE already trusts: dm-multipath for paths, shared thick LVM for the datastore, and pmxcfs for cluster state. It adds no on-disk format and no kernel code. If you remove pvesan, the datastores are still ordinary LVM storages.

**Goals, in priority order**

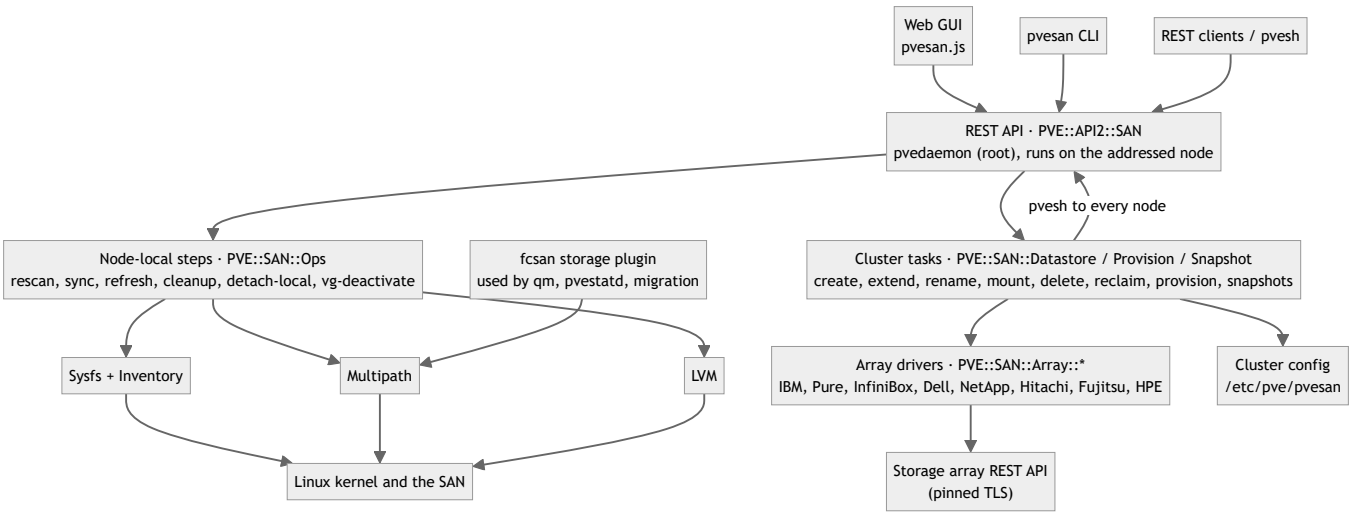
1. Never lose or corrupt data. Every destructive step checks usage on every node first, and LVM metadata changes only happen under Proxmox VE's own cluster storage lock.
2. Match the vCenter operator experience: the same verbs, run cluster-wide from any node, with a task log.
3. Stay upgrade-safe. Change as little of pve-manager as possible, and re-apply those changes automatically after an upgrade.
4. Keep GUI, CLI and API identical, by building the CLI and GUI on the API.

| Layer | ESXi / vCenter | Proxmox VE + pvesan |
|---|---|---|
| HBA and transport | vmhba, FC driver | `qla2xxx` / `lpfc`, `/sys/class/fc_host` |
| Device identity | `naa.` device ID | SCSI WWID (`3` + NAA hex), same value |
| Multipathing | NMP with SATP and PSP | dm-multipath; the policy maps onto `path_grouping_policy`, `prio` and `failback` |
| Device display name | Device display name | Multipath `alias` (`/dev/mapper/<alias>`), kept cluster-wide |
| Datastore file system | VMFS, a clustered FS with on-disk locks | Shared thick LVM VG; metadata serialized by the pmxcfs cluster lock |
| VM disk | VMDK file on VMFS | One LV per disk (raw, or a qcow2 volume chain on Proxmox VE 9) |
| Datastore registry | vCenter inventory | `/etc/pve/storage.cfg` entry of type `fcsan` |
| Cluster-wide config | vCenter DB, host profiles | `/etc/pve/pvesan/config.json` on pmxcfs |
| Rescan | `esxcli storage core adapter rescan` | sysfs `scan` and `rescan`, then a multipathd resize |
| Detached device | Device state `off` | WWID blacklisted, paths deleted, removed again after each rescan |
| Array plugin (Pure, InfiniBox, ...) | vSphere plugin | pvesan array drivers |

The largest difference is the datastore layer. VMFS lets many hosts allocate space at the same time using on-disk locks. Shared LVM has one metadata writer at a time, enforced with a cluster-wide lock, while data I/O to existing LVs runs in parallel on every node. So the design rule is: metadata changes (create, extend, resize, delete) go through the lock, data paths never do.

## Architecture

The GUI, the CLI and REST clients all call one API. A cluster task runs on the node it was called on and reaches every other node through that same API. The storage plugin is the only piece Proxmox VE itself calls, from `qm`, `pvestatd` and migration.




| Component | File | Integration point |
|---|---|---|
| Storage type `fcsan` | `PVE/Storage/Custom/FCSANPlugin.pm` | Proxmox VE loads every module in `Storage/Custom`. It subclasses `LVMPlugin` and overrides `activate_storage`, `free_image` (opt-in discard), `options` and `properties` |
| REST API | `PVE/API2/SAN.pm` | Registered as `/nodes/{node}/san` by one marked line in `PVE/API2/Nodes.pm` |
| CLI | `PVE/CLI/pvesan.pm`, `/usr/sbin/pvesan` | `PVE::CLIHandler` maps each command onto an API method |
| GUI | `/usr/share/pve-manager/js/pvesan.js` | One `<script>` line in `index.html.tpl`; an Ext `override` of `PVE.panel.Config` adds the datacenter panel below Datacenter → Storage and the node panel to the node's Disks group; the datacenter tables ask every online node in parallel (`GET /nodes/{node}/san/...`) and merge the answers in the browser |
| Integration | `/usr/share/pve-fcsan/pvesan-integrate` | Run by `postinst`, re-run by a dpkg trigger when pve-manager replaces either file |
| Health timer | `pvesan-monitor.timer`, `pvesan-alert-*.txt.hbs` | `pvesan monitor-run` every 5 minutes on every node; alerts through `PVE::Notify` |
| Boot sync | `pvesan-sync.service` | `pvesan sync` after `pve-cluster` and `multipathd`, before `pvestatd` and guests |

| Component | File | Integration point |
|---|---|---|
| Timers | `pvesan-snapshot.timer`, `pvesan-reclaim.timer` | `pvesan snapshot-run` every 15 minutes and `pvesan reclaim-run` weekly on every node; a claim in the cluster DB picks one node per job |
| Array drivers | `PVE/SAN/Array/*.pm` | Called only by provisioning workflows |

## Data model

State lives in three places, each with one owner; each node renders a fourth from them.

| Store | Location | Holds | Written by |
|---|---|---|---|
| Storage registry | `/etc/pve/storage.cfg` (pmxcfs) | `fcsan` entry: `vgname`, `shared`, `nodes`, `disable`, `fc-wwids`, `fc-autorescan`, `fc-discard` | Proxmox VE storage API (`PVE::API2::Storage::Config`) |
| pvesan cluster DB | `/etc/pve/pvesan/config.json` (pmxcfs) | Per WWID: `alias`, `policy`, `detached`, `datastore`. Per datastore: `label`, linked `array` and `volumes`; for a snapshot copy `copy_of` and `snapshot`; `snapshot_schedule`, `auto_reclaim` and job state. Per array: type, address, user, fingerprint, vendor settings. Per fabric (FC switch): type, address, user, fingerprint, `vsan`, `vf_id`, `zoneset` | `PVE::SAN::Config::update` under the `pvesan` cfs domain lock |
| Array secrets | `/etc/pve/priv/pvesan/<array>.pw` | Password or API token (switches: `fabric-<id>.pw`) | Root only |
| Monitor cache | `/etc/pve/pvesan/monitor.json` (pmxcfs) | Array views per datastore, array problems, lease, member list per array | `PVE::SAN::Monitor` under the `pvesan` domain lock |
| LVM metadata | On the LUNs | VG, PVs, LVs | LVM commands, only under `cluster_lock_storage` |
| Rendered, per node | `/etc/multipath/conf.d/pvesan.conf`, `/etc/multipath/wwids`, LVM devices file | Aliases, policies, blacklist of detached WWIDs, claimed WWIDs | `PVE::SAN::Ops::sync` |
| Node health state | `/var/lib/pvesan/health.json` | Highest path count seen per LUN, last reported problems | `PVE::SAN::Monitor` on that node |

**Why a separate cluster DB?** Aliases and path policies apply to LUNs, not storages: a LUN can have a policy before it belongs to a datastore, and a detached LUN belongs to none. `storage.cfg` has no place for that, so pvesan keeps a small JSON file next to it on pmxcfs.

**Rendering.** `render_conf()` is a pure function (device table in, multipath snippet out, sorted by WWID), so the same input always gives the same bytes. `sync` writes the file and runs `multipathd reconfigure` only when the content changed, which makes it cheap and safe to run at boot, during activation, and twice in every create.

```
multipaths {
    multipath {
        wwid 3600a098038303053453f463045727a41
        alias ds1
        path_grouping_policy group_by_prio
        prio alua
        path_selector "service-time 0"
        failback immediate
    }
}
```

**Identity.** Everything is keyed by WWID, never by `sdX` or `dm-N`, which change across reboots and differ between nodes. sysfs gives `naa.<hex>`; pvesan turns it into multipath's form, `3<hex>`. For vendor-specific (t10) IDs it calls `scsi_id`, the same source multipath uses.

**Usage classification.** For each LUN, `Inventory::devices` decides one usage, in this order: `datastore:<id>`, `lvm-vg:<vg>`, `lvm-pv`, `in-use:<holders>`, `signature:<type>` / `partitioned:<type>` (from blkid), `free`. Only `free` LUNs can be used without `force`; the first four are never wiped.

## Operation internals

Every cluster operation follows the same pattern:

1. Check the situation on every node through the API.
2. Change cluster state on pmxcfs.
3. Apply that state on every node.
4. Make the single LVM metadata change on the coordinating node, under the lock.

The fan-out (`PVE::SAN::Cluster::on_nodes`) calls `pvesh <verb> /nodes/<n>/san/<step>` for each node, the local one included, so every node runs the same code. It collects every node's failure before it aborts, so one task log shows all problems.

**Rescan (`Ops::rescan`).** Snapshot the SAN paths; optionally issue a LIP per port and wait for `Online`; write `- - -` to each FC/iSCSI host's `scan`; write `1` to each known path's `rescan` to re-read capacity; `udevadm settle`; delete paths of detached WWIDs; `multipathd resize map` for maps whose paths grew; return the difference.

**Sync (`Ops::sync`).** Render the snippet, write it if changed, and add or remove WWIDs in `/etc/multipath/wwids`. pvesan edits that file directly instead of calling `multipath -a`, which needs the LUN to be visible, and not every

node sees every LUN. If anything changed, run `multipathd reconfigure` and wait for the maps of LUNs this node has paths to; add them to the LVM devices file when it is enabled.

**Create (`Datastore::create`).**

1. Refuse if the storage ID or VG exists.
2. `check_visible`: every target node must see each WWID with an active path.
3. `check_unused_locally`: classify each LUN.
4. `register_devices`: alias in the cluster DB, `sync` on all nodes, wait for `/dev/mapper/<alias>`.
5. `wipefs` (only with `force`), `pvcreate`, `vgcreate --addtag pvesan`.
6. `PVE::API2::Storage::Config->create(type => 'fcsan', shared => 1, ...)`, the same call Proxmox VE's own Disks → LVM wizard makes.
7. Roll back with `vgremove`/`pvremove` if step 5 or 6 fails; finally `sync` everywhere again.

**Extend.** *Grow* runs `devices/{wwid}/refresh` on every node (re-read size, resize map) and aborts if the nodes disagree; only then `pvresize` under `cluster_lock_storage`. *Add extent* reuses steps 2–5 of create, then `vgextend` under the lock and updates `fc-wwids`.

**Rename.** A label rename touches only the cluster DB. An ID rename requires every node online and no running guest that refers to the storage, deactivates the VG everywhere, moves the `storage.cfg` section (and optionally runs `vgrename`) inside `lock_storage_config`, then rewrites `^key: <id>:` references in every guest config, snapshot sections included.

**Unmount and mount** map onto Proxmox VE's two switches: removing nodes from `nodes`, or `disable 1`. The VG is deactivated on the affected nodes first, and `vg-deactivate` refuses if any LV is open.

**Delete** refuses while guest configs refer to the storage or LVs remain (without `destroy-data`), deactivates everywhere, deletes the storage through the API, runs `vgremove`/`pvremove`/`wipefs`, cleans the cluster DB and optionally detaches the LUNs.

**Detach** asks every node whether the LUN is in use, marks it `detached` (which blacklists and unclaims it on `sync`), then runs `detach-local` on every node to flush the map and delete each path. Every later rescan deletes the paths again.

**Reclaim** checks `discard_max_bytes > 0` on each backing map, creates `pvesan_reclaim` over the free space minus a reserve under the storage lock, releases the lock, runs `blkdiscard`, and removes the LV again under the lock.

**Activation (`FCSANPlugin::activate_storage`).** Before handing over to `LVMPlugin`, the plugin checks that each WWID in `fc-wwids` has a map. If one is missing and `fc-autorescan` is on, it runs a local rescan and sync, rate-limited to once per 5 minutes per storage by a stamp file in `/run/pvesan`. If the map is still missing, activation fails with a clear message and the storage shows inactive on that node only.

## Array integration

Array drivers turn pvesan from a SAN manager into a provisioning tool. The array becomes one more layer pvesan drives, ahead of the steps above.

| Module | Role |
|---|---|
| `PVE::SAN::Array` | Registry of array types to drivers; array settings in the cluster DB, secrets in `/etc/pve/priv/pvesan/`; certificate probe; collects every node's WWPNs and/or iSCSI initiator name (array setting `protocol`) |
| `PVE::SAN::Array::Base` | Shared driver code: TLS-pinned HTTP client, naming rules, vendor-neutral host and host-cluster registration |
| `PVE::SAN::Array::<Vendor>` | One module per platform |
| `PVE::SAN::Provision` | Workflows: register the cluster, provision, grow, destroy with the volume, link |
| `PVE::SAN::ISCSI` | Node-local open-iscsi steps: initiator name, sessions, discovery and login, portals recorded for boot |

**Driver contract.** A driver implements `system_info`, `pools`, `hosts`, `host_clusters`, `create_hostcluster`, `create_host`, `add_host_ports`, `add_to_hostcluster`, `create_volume`, `volume`, `find_volume_by_wwid`, `map_to_hostcluster`, `unmap_from_hostcluster`, `expand_volume` and `delete_volume`. A volume always comes back with its multipath WWID, which connects array objects to everything else in pvesan. Drivers whose arrays present volumes differently override `ensure_hostcluster`. Three hooks cover vendor differences: `options` (required settings such as `svm` or `portgroup`), `valid_hostcluster_name`, and `wwid_after_map`.

**Trust.** No connection is made without a pinned SHA-256 certificate fingerprint. `array-probe` reads it, the admin compares it with the array's own display, and from then on only that certificate is accepted. The pin is checked by pvesan's own TLS verify callback, which accepts the server certificate only when its fingerprint matches: CA trust and host names play no part, so a certificate signed by a trusted CA but not pinned is refused too. Connections are HTTPS only, redirects are not followed and replies are limited to 64 MiB. Switches are trusted the same way. Credentials are tested before anything is saved.

**Registration** (`ensure_hostcluster`) is idempotent. For each node, an array host that already owns any of its WWPNs is reused and missing ports are added; otherwise a host named after the node is created (with "force", because ports may not be logged in until zoning is done). Every host then joins the cluster's host cluster. Ports split across hosts, or a host already in another host cluster, are refused: fixing those is an admin decision on the array.

**iSCSI.** A node's ports are FC WWPNs (16 hex digits) or iSCSI names (`iqn.`/`eui.`, lower case). The array setting `protocol` (`fc` by default, `iscsi`, `both`) decides which ones `PVE::SAN::Array::node_ports` collects: WWPNs from each node's `adapters`, the initiator name from each node's `GET iscsi` (`/etc/iscsi/initiatorname.iscsi`; the sysfs `iscsi_host` only exists once a session does). Hosts carry `wwpns` and `iqns`, and `create_host` / `add_host_ports` take both lists. A driver opts in with `supports_iscsi` and an optional `target_portals` primitive (`[ { address, port, iqn } ]`); `ensure_hostcluster` refuses IQNs for any other driver ("iSCSI host registration is not supported yet") before it changes anything on the array, and `array add`/`set` refuses the setting. After an iSCSI registration, `Provision::iscsi_login` calls the node step `POST iscsi-login` on every node for each portal (the array's, given ones, or the ones recorded for the array). The node step runs `iscsiadm -m discovery -t sendtargets`, sets `node.startup=automatic` on every record and logs in only where `iscsiadm -m session` shows no session, so it is idempotent; it fails only if no session to that portal exists afterwards. It records the portal in `iscsi.portals` of the cluster config, and `pvesan-sync.service` runs `pvesan iscsi restore` at boot (before `pvesan sync`, so the new paths get multipath maps), which logs in to recorded portals without a session, including on nodes that joined later. A rescan then finds iSCSI LUNs exactly like FC ones (`scan` on the `iscsi_host`).

| Platform | iSCSI host registration | Target portals | Checked against |
|---|---|---|---|
| Pure FlashArray | HostPost `iqns`, HostPatch `add_iqns` | `GET ports`: `portal`, `iqn` | go-pure-client FA 2.50 docs |
| IBM Spectrum Virtualize | `mkhost -iscsiname`, `addhostport -iscsiname`; `lshost iscsi_name` | `lsnode iscsi_name`; `lsip` (portset0) or `lsportip` (host=yes) | Cinder `storwize_svc_common.py` |
| Dell PowerStore | initiators `{port_name, port_type: iSCSI}` | `ip_pool_address` with purpose `Storage_Iscsi_Target`, `ip_port(target_iqn)` | Cinder PowerStore client, PyPowerStore |
| INFINIDAT InfiniBox | host port `{type: iscsi, address}` | `network/spaces` with service `ISCSI_SERVICE`: enabled `ips`, `properties.iscsi_iqn`, `iscsi_tcp_port` | InfiniSDK, Cinder `infinidat.py` |
| NetApp ONTAP | igroup protocol `iscsi` (`mixed` for both; fixed at creation) | `network/ip/interfaces` with service `data_iscsi`; `protocols/san/iscsi/services` `target.name` | `na_ontap_igroup`, Cinder ONTAP REST client |
| NetApp E-Series | host port `{type: iscsi, port: IQN}` | `graph` hostside `ioInterface` of type iscsi; `graph/xpath-filter` `/storagePoolBundle/target` | SANtricity Ansible (`_host`, `_facts`, `_iscsi_target`) |
| Hitachi VSP | `host-iscsis {iscsiName, portId, hostGroupNumber}` in the host group on each iSCSI target port (`portType ISCSI`) | `ports/<id>` `ipv4Address`, `tcpPort`; host group `iscsiName` | Cinder `hbsd_rest_api.py`, `hbsd_rest_iscsi.py` |
| Dell PowerMax | `initiatorId` with IQNs | port group `symmetrixPortKey`, then `system/.../director/<d>/port/<p>` `ip_addresses`, `identifier` | PyU4V `provisioning.py`, `system.py` |
| Fujitsu ETERNUS | not supported yet (the RESTful API reference could not be checked for it) | | |

**Provision** runs inside one task: register the cluster; create the volume; map it; rescan every 30 s and poll every node every 10 s until each node has an active path (up to 120 s); record the path policy; run the normal datastore create; store the link `datastores.<id> = { array, volumes[] }`. If anything fails before the datastore exists, the task unmaps the volume, cleans dead paths on all nodes, deletes the volume and drops the device record.

| Platform | API | Checked against | How the cluster is presented | Volume ID → WWID |
|---|---|---|---|---|
| IBM Spectrum Virtualize | `POST /rest/v1/<cli-command>[/<obj>]`, token from `/auth` | IBM REST API reference | Host cluster | `3` + `vdisk_UID` |
| Pure FlashArray | `/api/2.x/...`, API token → `x-auth-token` | Pure's generated SDK docs (go-pure-client) | Host group; mapping = connection | `3624a9370` + serial |

| Platform | API | Checked against | How the cluster is presented | Volume ID → WWID |
|---|---|---|---|---|
| INFINIDAT InfiniBox | `/api/rest/...`, Basic auth, `approved=true` on changes | InfiniSDK source | Cluster; LUN on the cluster, unmapped by LUN number | `36` + serial |
| Dell PowerStore | `/api/rest/...`, `login_session` → `DELL-EMC-TOKEN` + cookie | PyPowerStore | Host group (created with its first host) | `wwn` |
| Dell PowerMax / VMAX | Unisphere `/univmax/restapi/<ver>/sloprovisioning/symmetrix/<id>` | PyU4V | Host group + storage group `<hc>_SG` + masking view `<hc>_MV`; volumes created in the SG | `effective_wwn` |
| NetApp ONTAP | `/api/...`, jobs awaited | `netapp.ontap` Ansible modules | One igroup (FCP, iSCSI or mixed); one FlexVol + LUN per datastore | `3600a0980` + hex(serial) |
| NetApp E-Series | `/devmgr/v2/storage-systems/<ssid>` | SANtricity Ansible collection | Host group (Linux DM-MP host type) | `wwn` |
| Hitachi VSP | `/ConfigurationManager/v1/objects/storages/<id>`, session token, jobs awaited | OpenStack Cinder Hitachi driver, Configuration Manager reference | One host group per target port; LU path on each | `naaId` (known after mapping) |
| Fujitsu ETERNUS DX S5 / AF S3 | `:5665/api/v1/...`, `POST session` → Bearer token, jobs awaited and deleted | ETERNUS RESTful API Reference (P3AG-5492) | One FC host per WWN (the API has no host groups); every volume connected to all of them with one host LUN | `3` + `uid` |
| Dell Unity XT | `/api/types/...`, `/api/instances/...`, Basic auth + `X-EMC-REST-CLIENT: true`; `EMC-CSRF-TOKEN` and session cookie from a GET for writes | storops (`unity/client.py`, `unity/resource/*.py`), OpenStack Cinder Unity driver | One host per node with its FC initiators (WWNN:WWPN); no host groups, so every LUN's host access lists all node hosts | `3` + `wwn` |
| HPE Alletra 5000 / 6000 (Nimble) | `:5392/v1/...`, `POST tokens {data: {username, password}}` → `X-Auth-Token`; bodies wrapped in `{data}`, asynchronous jobs awaited | HPE nimble-python-sdk, OpenStack Cinder Nimble driver | One FC initiator group; volume presented by an access control record | `2` + `serial_number` (EUI-64; HPE common-host-libs) |
| HPE Primera / Alletra 9000 / 3PAR | `/api/v1/...`, `POST credentials` → `X-HP3PAR-WSAPI-SessionKey`, session deleted afterwards | HPE python-3parclient, OpenStack Cinder 3PAR driver | Host set (persona 2, Generic-ALUA); one VLUN per volume to `set:<host set>` with an automatic LUN | `3` + `wwn` |

## Array snapshots and copies

`PVE::SAN::Snapshot` builds the datastore-level workflows on five optional driver primitives: `snapshot_create(\@vols, $name)` (one point in time for all volumes), `snapshots(\@vols)`, `snapshot_delete`, `snapshot_clone` (a new writable, unmapped volume) and `snapshot_restore` (overwrite a volume in place).

`supports_snapshots` and `supports_group_snapshots` say what a driver can do; the base class dies with "not supported yet" for the rest.

| Platform | Snapshot | Several volumes | Clone | Restore in place | Checked against |
|---|---|---|---|---|---|
| Pure FlashArray | `POST volume-snapshots?source_names=<vol> {suffix}` | Protection group `pvesan-<vol>`, `POST protection-group-snapshots` | `POST volumes {source}` | `POST volumes?overwrite=true {source}` | go-pure-client FA 2.50 docs |
| INFINIDAT InfiniBox | `POST volumes {name, parent_id, write_protected}` | `POST volumes/group_snapshot` | Writable child of the snapshot | `POST volumes/<id>/restore` | InfiniSDK (`dataset.py`, `volume.py`) |
| Dell PowerStore | `POST volume/<id>/snapshot {name}` | Write-order consistent volume group `pvesan-<vol>`, `POST volume_group/<id>/snapshot` | `POST volume/<snap>/clone` | `POST volume/<id>/restore`, or `POST volume_group/<id>/restore` for a set, both with `create_backup_snap` | PyPowerStore (`protection.py`, `provisioning.py`, `utils/constants.py`) |
| NetApp ONTAP | `POST storage/volumes/<flexvol>/snapshots {name}` (the LUN's own FlexVol) | Not yet (consistency groups) | FlexClone `POST storage/volumes {clone.parent_volume.name, clone.parent_snapshot.name, clone.is_flexclone}`; its `lun0` | `PATCH storage/volumes/<flexvol> {restore_to.snapshot.name}` | NetApp Ansible (`na_ontap_snapshot`, `na_ontap_volume`, `na_ontap_volume_clone`) |
| IBM Spectrum Virtualize | Thin target `pvs_<vol id>_<name>`, `mkfcmap -copyrate 0`, `prestartfcmap`, `startfcmap` | FlashCopy consistency group `pvs_<name>_<first id>` (`mkfcconsistgrp`, `startfcconsistgrp`) | Thin volume fed by `mkfcmap -copyrate <copy-rate> -autodelete` from the snapshot (default 100) | `mkfcmap -copyrate <copy-rate> -autodelete` snapshot → volume, `prestartfcmap -restore`, `startfcmap -restore` | OpenStack Cinder Storwize/SVC driver (`storwize_svc_common.py`) |
| Dell Unity XT | `POST types/snap/instances {storageResource, name}` | Consistency group `pvesan-<vol>` (`createConsistencyGroup`, `modifyConsistencyGroup {lunAdd}`), snapshot of the group; member LUNs are changed through `lunModify` | Thin clone `createLunThinClone {name, snap}` (thin LUNs; not for group members) | `POST instances/snap/<id>/action/restore`, the backup snapshot deleted afterwards | storops (`snap.py`, `cg.py`, `thinclone_helper.py`), Cinder Unity `client.py` |

| Platform | Snapshot | Several volumes | Clone | Restore in place | Checked against |
|---|---|---|---|---|---|
| HPE Alletra 5000 / 6000 (Nimble) | `POST snapshots {name, vol_id}` | Volume collection `pvesan-<vol>` (`PUT volumes/<id> {volcoll_id}`), `POST snapshot_collections {name, volcoll_id}`, deleted as a whole | `POST volumes {clone: true, base_snap_id, inherit_acl: false}` | Volume offline, `POST volumes/<id>/actions/restore {base_snap_id}`, online again | nimble-python-sdk (`volumes.py`, `snapshots.py`, `volume_collections.py`, `snapshot_collections.py`), Cinder Nimble driver |
| HPE Primera / Alletra 9000 / 3PAR | Virtual copy: `POST volumes/<vol> {action: createSnapshot, parameters: {name, comment}}`, the pvesan name in the comment | Volume set `pvesan-<vol>`, `POST volumesets/<set> {action: createSnapshot}` with `<base>-@count@` | Online physical copy `POST volumes/<snap> {action: createPhysicalCopy, parameters: {destVolume, destCPG, online}}` | `PUT volumes/<snap> {action: 4}` (promote virtual copy, online), task awaited | python-3parclient (`client.py`), OpenStack Cinder (`hpe_3par_common.py`) |

A **snapshot set** is every array snapshot with the same pvesan name across the datastore's volumes (`group_sets`); a set missing a volume is listed as incomplete and can't be mounted or rolled back. A group snapshot (Pure protection group) is deleted once, not per member.

**Mount as copy.** Clone each member, map the clones, `wait_visible`, register the devices under the copy's ID (aliases, multipath), then `vgimportclone --basevgname san_<copy>` on the coordinating node, which writes new PV and VG UUIDs so LVM never sees two VGs with the same identity for longer than the import. Every LV is set read-only (`lvchange --permission r`) and the copy is registered as an `fcsan` storage with content `images`. The cluster DB records `{ array, volumes, copy_of, snapshot }`, so remove-copy is the normal delete-with-volume. A failure at any step unmaps and deletes the clones again.

**Restore a VM.** The VM's current config gives the disks on the datastore; each one present in the copy is copied with `qemu-img convert -n` (raw over the live LV with the VM stopped, or converted to raw into newly allocated volumes for a new VM ID). A disk that is a qcow2 volume-chain layer with a backing file in the copy is refused, because it can't be restored on its own. `new_vm_config` (a pure function, unit tested) rewrites the config for a new VM: disks remapped, other storages dropped, new MACs, SMBIOS UUID and vmgenid, links down.

**Roll back.** Requires a typed confirmation, every guest stopped, every node of the datastore online and no mounted copy. The storage is disabled, the VG deactivated on all nodes, each volume restored by the array, every node re-reads the LUNs, and the storage is enabled again.

**Schedules.** The cluster DB holds `snapshot_schedule = { interval, keep }`, `auto_reclaim` and per-job state `{ last, error, claimed }`. `pvesan-snapshot.timer` (every 15 minutes) and `pvesan-reclaim.timer` (weekly) fire on every node. Each run reads the config without a lock and returns at once if nothing is due; otherwise it claims the due jobs under the `pvesan` domain lock (a claim older than 6 hours is taken over, in case a node died mid-run), runs them outside the lock and records the result. Retention removes only `auto-<date>-<time>` snapshots beyond `keep`, never manual or mounted ones.

## Per-disk array volumes (`fcsanvol`, experimental)

> **Experimental (since 1.5.0):** Pure FlashArray only, raw VM disks only, and not yet run on a real array.

`PVE::Storage::Custom::FCSANVolPlugin` is a second storage type that maps the Proxmox storage plugin API onto array volumes; `PVE::SAN::Volume` holds the logic so it is unit tested without Proxmox.

- **Identity without an array call.** The multipath alias of a disk is `<storage>-<volname>`, so `path()` (called on every VM start and by `qemu_blockdev_options`) is `/dev/mapper/<alias>` and needs neither the array nor the cluster DB. The array volume is `<storage prefix>-<volname>`, so `list_images` needs only the array. The cluster DB keeps per WWID `{ alias, datastore, disk, lun, array_volume }`; there is no LVM layer and no metadata on the LUNs.
- **Targeted rescan.** `alloc_image` creates, maps and records the volume with its LUN number but scans nothing. `activate_volume` on the node that runs the VM writes `- - <lun>` to each FC/iSCSI host's `scan` (`Ops::rescan_lun`), runs the normal `sync` and waits for the map. This keeps rescans cheap on nodes with hundreds of LUNs.
- **Scale guard.** `fc-max-luns` (default 255) refuses new disks before LUN-per-host limits (for example 256 host LUNs without flat addressing on some arrays) are reached.
- **Snapshots and clones** reuse the phase 2 primitives (`snapshot_create`, `snapshots`, `snapshot_delete`, `snapshot_restore`, `snapshot_clone`) plus `copy_volume`, `rename_volume`, `host_lun`, `volumes_named` and `volume_by_name` (`supports_disk_volumes`). On Pure: `GET volumes?destroyed=false` / `filter=name='...'`, `GET connections` → `lun`, `PATCH volumes {name}`, `POST volumes {source}` (go-pure-client FA 2.50 VolumesAPI, VolumePatch, ConnectionsAPI, Connection).
- **Delete** marks the WWID detached cluster-wide and removes its paths on every node before the volume is disconnected, so no node is left with dead paths.
- **Status** (asked by `pvestatd` every 10 s on every node) is the configured pool's capacity, cached for 60 s per node in `/run/pvesan`.
- **Safety.** A LUN that is an `fcsanvol` disk shows as `vm-disk:<storage>:<disk>` in the device list and can't be used for a datastore or detached by hand.

## Replication and disaster recovery (experimental)

> **Experimental:** replication setup on Pure FlashArray only; not yet run on real arrays. `PVE::SAN::DR` holds the logic.

**Driver primitives** (`supports_replication`): `replication_partners`, `replication_enable(vols, partner, interval, retain)`, `replication_disable`, `replicate_now(vols, name)`, `replication_status(vols)` on the source array; `replicas`, `replica_snapshots(replica)` and `replica_clone(member, name)` on the DR array. Other drivers report "replication is not supported yet".

| Step | Pure FlashArray call (go-pure-client FA 2.50; target naming as in Pure's Ansible `purefa_pgsnap`) |
|---|---|
| Partners | `GET array-connections` → `remote.name`, `status`, `type = async-replication` |

| Step | Pure FlashArray call (go-pure-client FA 2.50; target naming as in Pure's Ansible `purefa_pgsnap`) |
|---|---|
| Enable | Protection group `pvesan-<first volume>` with the volumes, `POST protection-groups/targets?group_names=<pg>&member_names=<partner>`, `PATCH protection-groups {replication_schedule: {enabled, frequency (ms)}, target_retention: {all_for_sec}}` |
| Replicate now | `POST protection-group-snapshots?source_names=<pg>&replicate_now=true {suffix}` |
| Status | `GET protection-group-snapshots/transfer?source_names=<pg>` → `started`, `completed`, `progress`, `data_transferred` |
| Replicas (DR array) | `GET protection-groups` with `is_local = false`, named `<source array>:pvesan-*` |
| Replica snapshots | Transfers with `completed` set, and the volume snapshots `<source>:<pg>.<suffix>.<volume>` |
| Copy for the DR site | `POST volumes?names=<new> {source: {name: <volume snapshot>}}` |

**Configs travel with the data.** `/etc/pve` stays at the primary site, so each protected datastore carries its guests: LV `pvesan-dr-meta` (8 MiB) on the datastore's VG holds two 4 MiB slots. Each slot is `PVESANDR1`, a JSON header `{gen, len, sha256}` and a JSON payload `{storage, written, guests: [{vmid, type, node, name, snapshots, config}], plan}`. A write goes to the older slot with the next generation, under the storage's cluster lock, followed by `fsync`; a reader takes the newest slot whose checksum matches, so a replica taken in the middle of a write still has the previous copy. `pvesan-dr.timer` (every 15 minutes, claim kind `drmeta` like the snapshot schedules) and every replicate-now write it. Only the current section of each config is stored; snapshot sections are dropped.

**Bringing a replica up** (`from_replica` → `import_luns`): copy each member of the newest complete (or the named) replica snapshot to a new volume `<storage>-<timestamp>[-NN]`, map it to the host cluster, wait until every node sees it, register the multipath aliases, `vgimportclone --basevgname san_<storage>` (new PV and VG UUIDs), read the metadata, check the VM IDs against the cluster (`vmid-offset` moves them), `lvrename` disks whose VM ID changes, register an `fcsan` storage and write each guest's config on the node that runs the task. Any error before the storage is registered deactivates the VG, unmaps and deletes the copies and forgets the devices.

**Test versus failover.** `dr_guest_config` rewrites the disks on the replicated datastore to the new storage ID and volume names, keeps disks on storages that exist at the DR site, empties CD-ROMs whose storage is missing and drops `parent`, `lock`, `vmstate` and the `running*` keys. A test copy also gets new MACs (Proxmox's OUI) and SMBIOS UUID, a new `vmgenid`, `-drtest` in its name, `link_down=1` on every NIC and no `onboot`. A failover keeps the identity. The datastore record keeps `dr_import {mode, source, snapshot, node, guests}`; `cleanup` refuses anything that is not a test.

**Planned failover** (`prepare_failover`): shut guests down in reverse plan order (guests outside the plan first), refuse while any still runs, write the metadata, unmount the datastore cluster-wide (VG deactivated on every node), send `dr-final-<date>-<time>` and wait until the partner reports it complete.

### Automatic FC zoning

| Module | Role |
|---|---|
| `PVE::SAN::Fabric` | Registry of FC switches (one per fabric, VSAN or virtual fabric) in the cluster DB under `fabrics`; passwords in `/etc/pve/priv/pvesan/fabric-<id>.pw`; same certificate pinning as arrays |
| `PVE::SAN::Fabric::Base` | Switch driver contract and shared HTTPS client |
| `PVE::SAN::Fabric::Brocade` | FOS REST API (8.2.1+): one session per call (`/rest/login`, `/rest/logout`), `brocade-name-server`, `brocade-zone` defined and effective configuration |
| `PVE::SAN::Fabric::Cisco` | MDS NX-API `ins_api`: `show fcns database detail`, `show zone`, `show zoneset [active]`, `show zone status`, zoning config commands |
| `PVE::SAN::Zoning` | Plan and apply single-initiator zones between the nodes and an array |

**Driver contract.** A switch driver implements `system_info`, `name_server` (every logged-in N_Port: WWPN, WWNN, role, symbolic name), `zoning` (active zone set, defined zone sets with their zones, defined zones and active zones with aliases resolved to WWPNs, and whether another change is open) and `apply_zones($zoneset, \@zones)`. An array driver opts in by reporting its FC ports with `target_ports`: IBM `lstargetportfc` rows with `host_io_permitted=yes` (the NPIV ports when there are any), InfiniBox `components/nodes` FC ports that are up (as Cinder's `infinidat.py`), Pure `ports`, HPE WSAPI `ports`.

**Planning** (`Zoning::compute`, pure and unit tested). For each fabric, the node ports and array ports in its name server are the ones to pair. For each node port the zone `pve_<host cluster>_<node>_<WWPN>_<array>` should hold that port and every array port on the same fabric. If an active zone (any name) already holds all of them, nothing changes. If the pvesan zone exists, the missing members are added; otherwise it is created. The zone set is the fabric's `zoneset` setting or the active one. A fabric is blocked when another change is open, when no zone set is known, or when the defined zone set differs from the active one (activating would also push someone else's saved changes). The plan also reports node ports and array ports on no fabric, and nodes reached by only one fabric.

**Applying** recomputes the plan from live data, refuses any blocked fabric before it touches any, then per fabric creates or extends the zones, adds new zones to the zone set and activates it. Brocade: zone and cfg `POST`s open a transaction; `PATCH effective-configuration/cfg-name/<cfg>` with the current checksum saves and enables it; on any error `cfg-action/4` aborts the transaction. Cisco: one `cli_conf` batch (`zone name`, `member pwwn`, `zoneset name`, `member`, `zoneset activate`, and `zone commit` in enhanced mode); on error `no zone commit` discards an enhanced-mode session; the running configuration is then saved. pvesan never removes or renames zones.

## Consistency, locking, safety, permissions

The one invariant: **only one node at a time writes LVM metadata for a shared VG, and every node sees the same device size before LVM is told about it.**

| Lock | Taken through | Guards | Held for |
|---|---|---|---|
| Cluster storage lock | `PVE::Storage::Plugin->cluster_lock_storage($id, 1, 60, ...)` | `pvcreate`/`vgextend`, `pvresize`, reclaim LV create/remove; Proxmox VE's own volume allocation takes it too | Seconds |

| Lock | Taken through | Guards | Held for |
|---|---|---|---|
| Monitor lease | A `{ node, time }` entry in `monitor.json`, written under the `pvesan` domain lock | Which node reads the arrays this run | One timer interval |
| storage.cfg lock | `PVE::Storage::lock_storage_config` | ID rename (section move plus `vgrename`) | Seconds |
| pvesan domain lock | `cfs_lock_domain('pvesan')` | Read-modify-write of `config.json`, and the claim of scheduled snapshot and reclaim jobs | Milliseconds |

pmxcfs aborts code holding a lock for more than 60 s, so every slow step (node fan-out, waiting for maps, `blkdiscard`, array calls) stays outside locks.

**Safety rails**

- Visibility gate: create and extend abort unless every target node sees each LUN; grow aborts if nodes report different sizes.
- Usage gate: LUNs in use as a datastore, VG, PV or with holders are never wiped; signatures and partitions need explicit `force`.
- Reference gate: delete, ID rename and unmount refuse while guest configs refer to the storage or those guests run on the target nodes.
- Open-volume gate: `vg-deactivate` refuses while any LV is open.
- Rollback: a failed create removes the VG/PVs it made; a failed provision deletes the new array volume.
- Typed confirmation for Delete and Detach in the GUI.
- Integration guard: the line in `Nodes.pm` loads pve-fcsan inside an `eval`, so a pve-fcsan that fails to load disables only `/nodes/{node}/san` (with a warning in the journal), never the rest of the API. `pvesan-integrate` runs `perl -c` on the patched `Nodes.pm` and reverts if it fails. The GUI override catches its own errors, so a broken `pvesan.js` loses only the SAN panel.
- Uninstall guard: `prerm` refuses while `fcsan` storages exist.

**Permissions.** API calls are `protected` (handled by pvedaemon as root) and `proxyto => 'node'`. Reads need `Sys.Audit` on the node; node-local changes `Sys.Modify` on the node; anything cluster-wide or array-related `Sys.Modify` on `/` and `Datastore.Allocate` on `/storage`. Those privileges allow root-level storage changes on every node, the same as Proxmox VE's own storage management.

## Monitoring and health

`PVE::SAN::Monitor` gives the vSphere plugins' storage health view inside Proxmox VE, without a separate daemon.

**One timer, two roles.** `pvesan-monitor.timer` runs `monitor-run` on every node every 5 minutes. Every node checks its own paths. Then the node takes a lease in `/etc/pve/pvesan/monitor.json` (under the `pvesan` domain lock; a lease older than 4 minutes is taken over) and, if it got it, reads every array-linked datastore's array view, registers new nodes and evaluates array-level problems. Only one node talks to the arrays per run.

**Array view.** `collect_view` asks the driver for `volume_space` (provisioned, written, reduction or capacity saved), `pools` (filtered to the pools of the datastore's volumes) and `health`. Every driver answers provisioned size and pools through its existing primitives; written space, reduction and health come from drivers that implement them:

| Primitive | Pure FlashArray | INFINIDAT InfiniBox | Dell PowerStore |
|---|---|---|---|
| `volume_space` | `GET volumes?names=` → `space.virtual`, `space.data_reduction` | `GET volumes/<id>` → `used`, `capacity_savings` | `GET volume/<id>?select=id,size,logical_used` (falls back without `logical_used` before PowerStoreOS 3) |
| `health` | `GET alerts?filter=state='open'` | — | — |
| `target_ports` | `GET ports` → `wwn` | — | — |
| `remove_host` | `DELETE host-groups/hosts`, `DELETE hosts` | `DELETE clusters/<id>/hosts/<host>`, `DELETE hosts/<id>` | `PATCH host_group/<id> {remove_host_ids}`, `DELETE host/<id>` |
| Checked against | go-pure-client FA 2.50 docs (VolumeSpaceCommon, Alert, Port, HostGroupsAPI, HostsAPI) | InfiniSDK (`dataset.py`, `host_cluster.py`) | PyPowerStore (`utils/constants.py`, `provisioning.py`) |

The view is cached in `monitor.json` with its time; the API serves the cache when it is under 15 minutes old.

**Path health.** `path_problems` is a pure function: the node's devices, the LUNs it must see (`watched_luns`: LUNs of enabled datastores whose node list includes the node, detached ones excluded) and the highest path count seen per LUN (kept in `/var/lib/pvesan/health.json`). A LUN is missing, dead (no active path) or degraded (fewer active paths than ever seen). Learning the maximum catches paths that disappeared entirely after `dev_loss_tmo`, which a plain "active < total" check misses; `path-health-reset` forgets it.

**Alerts.** Each run compares the problem set with the previous one (node problems in the node's state file, array problems in `monitor.json`) and sends only the difference: new problems as `warning`, resolved ones as `info`, through `PVE::Notify::notify` with the `pvesan-alert` templates and the field `type=pvesan`, so the admin's matchers and targets decide where they go. A Proxmox VE without the notification system only gets the journal line.

**I/O rates.** `iostats` reads `/sys/block/dm-N/stat` of every multipath map twice, `interval` seconds apart, and derives IOPS, bytes per second, average latency per I/O (`ms` delta / `ios` delta) and utilisation (`io_ticks`). No device-mapper statistics regions are created, so nothing is left behind on the node.

**Relations.** `datastore_guests` reads every guest config on pmxcfs (`guest_refs`), so guests on offline nodes are listed too; `guest_storage` maps each disk of one guest to its datastore, VG, LUN aliases and array volumes from `storage.cfg` and the cluster DB.

**Node lifecycle.** `unregistered_nodes` (none of a node's WWPNs belongs to any array host) runs only when the cluster's member list differs from the one stored per array, so a stable cluster costs no array calls; the registration itself is the normal idempotent `ensure_hostcluster`, limited to the new nodes. `stale_hosts` marks hosts in the cluster's host cluster that are not named after a member and own none of the online nodes' ports; removal is refused while any node is offline, because an offline node's host would look the same.

**Zoning hints.** `zoning_rows` compares, per node and adapter, the online FC target logins (`fc_remote_ports`) with the array's ports: from `target_ports` where the driver has it, otherwise the union of the target ports any node uses for the array's LUNs.

**no_path_retry.** The device record can carry `no_path_retry`, rendered into the LUN's `multipath {}` section after the policy settings. Drivers whose arrays multipath-tools queues forever (`IBM 2145`, `NETAPP LUN`, `NFINIDAT InfiniBox`) return 12 from `default_no_path_retry`, which provisioning records for new LUNs.

## Development and testing

### Design decisions

| Decision | Chosen | Rejected, and why |
|---|---|---|
| Datastore format | Shared thick LVM (`fcsan` subclasses `LVMPlugin`) | GFS2/OCFS2: extra cluster stack, not supported by Proxmox. LVM-thin: not safe shared |
| Storage type | Custom `fcsan` type | Plain `lvm`: no LUN binding or auto-rescan (plain `lvm` storages on SAN are still listed and managed) |
| API exposure | One line in `Nodes.pm`, re-applied by a dpkg trigger | A separate daemon: second auth path and port. Runtime monkey-patching: fragile load order |
| GUI | Ext `override` of `PVE.panel.Config` | Patching `pvemanagerlib.js`: replaced on every update |
| Cluster fan-out | `pvesh` to each node, including the local one | Direct local calls: two code paths to test |
| Claiming LUNs | Edit `/etc/multipath/wwids` directly | `multipath -a`: fails on nodes that don't see the LUN |
| Plugin API version | Return the running `APIVER` | Pinning a range: a release for every Proxmox VE bump |
| Array drivers | Thin per-vendor modules on a shared base | One generic "SMI-S" driver: patchy vendor support, heavier dependencies |

**Layers.** `Util`, `Sysfs`, `Multipath` (parsing/rendering) and `LVM` (parsing) are pure Perl and read sysfs through `PVESAN_SYSFS_ROOT`, so a fixture tree replaces `/sys`. `Inventory`, `Ops`, `Config` and `Cluster` add Proxmox VE and command execution; `Datastore` and `Provision` compose them; `API2::SAN` exposes them; the CLI and GUI sit only on the API.

**Tests** (`make test`, 1371 checks):

| File | Covers |
|---|---|
| `t/10-util.t` | WWN/WWID normalization, aliases |
| `t/20-sysfs.t` | Discovery against a fake sysfs tree: FC and iSCSI hosts, paths, maps, ALUA standby, degraded LUNs, grown LUNs |
| `t/30-multipath.t` | Path-state parsing, golden multipath config render |

| File | Covers |
| --- | --- |
| `t/40-lvm.t` | LVM JSON reports |
| `t/50-cluster.t` | The exact `pvesh` command lines of the fan-out |
| `t/60…66-*.t`, `t/67-snapshots.t` | Every array driver against a fake array that keeps state (`t/lib/FakeArray.pm`): auth, request bodies, registration, full volume lifecycle, error formats |
| `t/68-snapshot-workflow.t` | Snapshot sets, retention, due jobs, VM config rewriting, and the snapshot, mount-copy, rollback and schedule workflows against a fake driver with the cluster steps stubbed |
| `t/69-monitor.t` | I/O rates, path problems and learned path counts, alert differences, pool and health problems, zoning hints, new and stale hosts, `no_path_retry` rendering, the monitoring primitives of Pure, InfiniBox and PowerStore, and the monitor run (lease, registration, alerts sent once) |
| `t/70-volumes.t` | `fcsanvol`: names and aliases, Pure's per-disk primitives, allocation with rollback on failure, the LUN limit, targeted activation, resize, snapshots, rollback, templates, linked clones, delete, and the plugin's features |
| `t/71-dr.t` | DR metadata slots (checksum, generations, torn slots), recovery plans, VM ID offsets and disk renames, guest configs for test and failover (VMs and containers), the metadata timer job, Pure's replication primitives against a fake array, and the replicate, planned-failover, test-failover, cleanup, failover and manual-import workflows against a fake driver |
| `t/73-hpe3par.t` | HPE Primera / Alletra 9000 / 3PAR against a stateful fake WSAPI: session key, re-login and logout, error format, CPG capacity, host set registration with persona and ports, host set VLUNs, the full volume lifecycle, virtual copies and volume set snapshots (the array's own copies hidden), online copies, promote with its task, and volume set clean-up |
| `t/72-snapshots-more.t` | Snapshots on NetApp ONTAP (FlexVol snapshots, FlexClone copies, restore, its own snapshots hidden, layout and serial checks) and IBM Spectrum Virtualize (FlashCopy targets, consistency groups, background copies, restore, clean-up after a failure) against stateful fake arrays |
| `t/75-unity.t` | Dell Unity XT against a stateful fake array: CSRF token and session cookie, request bodies, one host per node with WWNN:WWPN initiators, the volume lifecycle, host access lists, snapshots, thin clones, restore, consistency groups and the error format |
| `t/75-iscsi.t` | iSCSI names, the node's initiator name, portal parsing, the exact `iscsiadm` discovery, `node.startup` and login commands (idempotent, partial failures), boot restore, node ports per protocol, refusal by drivers without iSCSI, and each driver's iSCSI host bodies and target portals against fake arrays |
| `t/76-nimble.t` | HPE Alletra 5000/6000 (Nimble) against a stateful fake NimOS array: token login and re-login, `{data}` request bodies, paged lists, asynchronous jobs, the FC initiator group, WWID from the serial, the volume lifecycle (offline before delete), snapshots, volume and snapshot collections, clones, restore, error formats |
| `t/77-zoning.t` | Automatic zoning: the plan (zones created, extended or left alone, names, members per fabric, zone set membership, blocked fabrics, unplaced ports, redundancy), Brocade FOS REST (session, vf-id, name server, aliases, cfg enable with checksum, abort on error) and Cisco NX-API (fcns and zone parsing, enhanced-mode commit and discard, open session) against fake switches, IBM and InfiniBox FC target ports |
| `t/78-security.t` | The certificate pin (only the pinned certificate passes, also against a CA-signed one in a real TLS handshake), HTTPS only, no redirects, the reply size limit, malformed guest records and hookscripts in DR metadata, and that every GUI grid column is HTML-encoded |

| File | Covers |
| --- | --- |
| `t/79-gui.t` | GUI layout: the Datacenter → Storage entry and its tabs, the node panel's local tabs, that every endpoint the datacenter view asks each node for exists, and how answers and unreachable nodes are collected |
| `t/74-hardening.t` | `fc-discard` handing off to the LVM plugin's background discard (and the inline fallback), container configs and restore from a snapshot copy |
| `t/80-integrate.t` | `pvesan-integrate` against a fake pve-manager tree: insertion point, upgrade from an older line, refusal without the anchor, exact removal, and that the patched `Nodes.pm` still loads with a broken pve-fcsan |
| `t/90-compile.t` | Loads every module; fails on compile warnings. Uses the real Proxmox VE libraries when present, stubs otherwise |

The build also checks that every Proxmox function the code calls exists in the installed libraries, that the REST router accepts every API path, that `pvesan.js` parses, and that every module is in the package. CI runs the tests on every push.

**Validation so far.** The plugin runs on a 6-node Proxmox VE 9.2 cluster with QLogic and Emulex HBAs: install, integration, services, storage plugin, API, CLI, GUI asset, local and cluster rescan, refusal of invisible LUNs and CLI exit codes are verified there, and every array driver's first call was checked over pinned TLS. That lab work found and fixed seven bugs that unit tests could not see (an API path clash, a non-existent helper call, arguments lost in the cluster fan-out, a packaging omission, among others). Array drivers have not yet run against real arrays.

**Release steps**

1. Update `debian/changelog` and `CHANGELOG.md`.
2. `make test` and `make deb` on a clean Debian 12/13 host (CI does this on every tag).
3. Run the cluster acceptance checklist below on a lab cluster.
4. Tag `vX.Y.Z`; CI attaches the `.deb` to the GitHub release and rebuilds the signed apt repository on GitHub Pages from every release's `.deb` (`tools/build-apt-repo.sh`, key in the `APT_SIGNING_KEY` secret).
5. Roll out node by node, checking `pvesan-integrate status` after each.

**Cluster acceptance checklist**

- [ ] Present a LUN, `rescan-all`, confirm it shows `free` on every node
- [ ] Create a datastore and live-migrate a VM between all nodes
- [ ] Grow the LUN, `extend`, check the size in `pvesm status` on every node
- [ ] Add an extent, reboot one node, check the datastore comes back `active`
- [ ] Set policy `alua`, check `multipath -ll`, pull a fabric cable during I/O
- [ ] Rename the label online; rename the ID with guests stopped and boot them
- [ ] Unmount from one node, check migration to it is refused, mount again
- [ ] Reclaim on a thin LUN and confirm space returns on the array
- [ ] Delete with `--detach`, unmap, rescan, confirm no dead paths remain
- [ ] For each array driver: add, register, provision, grow, delete with volume
- [ ] For each driver with snapshots: snapshot, mount a copy, restore a VM over its disks and into a new ID, remove the copy, roll back a test datastore
- [ ] Upgrade pve-manager and confirm `pvesan-integrate status`

## Limitations, known risks, roadmap

### Known risks

| Risk | Why it might bite | How to check |
|---|---|---|
| Array API differences by code level | Drivers were checked against vendor SDKs, not yet against real arrays | Provision and delete a test datastore per array type; report results |
| Code in pve-storage or qemu-server that tests `type eq 'lvm'` | An `fcsan` storage could miss an LVM-specific branch | Snapshot, clone and migrate a VM on `fcsan`; fall back by changing the type to `lvm` |
| `Nodes.pm` anchor text changes | The integration can't find its insertion point | `pvesan-integrate apply` reports it and changes nothing |
| Widget-toolkit API drift | GUI classes could change in a new Proxmox VE | Open the panel after upgrades; check the browser console |
| DR replicas never brought up on real arrays | Pure's replication calls are checked against Pure's SDK docs only; target-side naming follows Pure's Ansible module | Run a test failover with a test datastore before relying on it |
| Switch drivers never run against real switches | Brocade and Cisco calls follow the vendors' API references and Broadcom's Ansible collection; NX-API show output field names vary by NX-OS release | `pvesan array zone-plan` first (read-only), then zone a test array on one fabric with a zoning backup |

### Functional limits compared with vCenter

- Thick LVM only at the datastore layer (the array can still be thin). Per-VM snapshots on shared storage need Proxmox VE 9 volume chains.
- Array snapshots, copies and restores need a driver with snapshot support (Pure, InfiniBox, PowerStore, ONTAP, IBM Spectrum Virtualize, HPE 3PAR); for other arrays snapshot or replica LUNs are imported by hand (`vgimportclone` procedure in the runbook). Snapshots are crash-consistent; VMs are not frozen.
- No Storage I/O Control, datastore clusters, Storage DRS or datastore browser. Datastores can't be shrunk.
- Nodes offline during a change catch up at boot through `pvesan-sync.service`.

### Roadmap

1. Real-array validation for every driver; fix what it finds.
2. A datacenter-level GUI view: all nodes' paths for one LUN side by side.
3. `datastore adopt`: convert existing `lvm` storages on SAN to `fcsan`.
4. Array snapshots for the remaining drivers (PowerMax, E-Series, Hitachi, ETERNUS); ONTAP consistency groups; container restore.
5. Per-LUN path statistics and fabric events.
6. Replication on InfiniBox and PowerStore; failback in one step; stretched clusters on active-active arrays.